Evaluation runs: 2pc-ds (two-pc-disel)

generated 2026-11-08 from brain-data 34d8e8fae 2026-11-08 19:23:20 +0100

Every campaign run on this benchmark, one row per instance (variant, specification, communication model, size) and engine; the newest run of an instance wins. Expected verdicts (the rule file): no entry in the rule file (the README states the expectation). Blue rows are at the current pins (TraceForge 367cdc6, whose predecessor 9a0629e is equivalent for every engine but the join; merge join 367cdc6; JMC 3c43b2d2): a rerun gives the same result. Red rows are superseded (an older pin, or the in-memory join that the merge join replaces) and their rerun is pending. A planned instance without a measurement is not a row: it is listed under "Not run". G_{ref}, G_{cand} = the graphs of the reference and of the implementation (the stateless engine's own count of specification graphs explored, repeats included, is G_{ref}^{expl}); G_x = the implementation graphs no specification graph covers (violating pairs); T = wall seconds; M = peak resident MiB; tier = the run's time and memory limits; * marks a verdict other than the expected one; 🕓 = the time tier was reached; oom = the memory limit was reached.

stateless

stateless: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	sweeps	G_{ref}^{expl}	run
2	400	256	conforms	0.2	2	1 h, 250 GB	16	2384	brain03, 2026-11-02
3	104976	82944	conforms	212.6	4	1 h, 250 GB	64	2574748	brain03, 2026-11-02

stateless: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	sweeps	G_{ref}^{expl}	run
2	400	256	conforms	0.2	2	1 h, 250 GB	16	2370	brain03, 2026-11-02
3	104976	82944	conforms	208.7	2	1 h, 250 GB	64	2559952	brain03, 2026-11-02
4	95883264		🕓 (1 h)			1 h, 250 GB			brain03, 2026-11-02

stateless: pair **skip-ack-loop**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	sweeps	G_{ref}^{expl}	run
2	400	64	64	violation	1.3	2	1 h, 250 GB	64	25600	brain03, 2026-11-02
3	104976			🕓 (1 h)			1 h, 250 GB			brain03, 2026-11-02

stateless: pair **skip-ack-loop**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	sweeps	G_{ref}^{expl}	run
2	400	64	64	violation	1.3	4	1 h, 250 GB	64	25600	brain03, 2026-11-02
3	104976			🕓 (1 h)			1 h, 250 GB			brain03, 2026-11-02

Not run (stateless). planned but never scheduled (grid points beyond the ladder): 4 bag correct.t16.p.r3, 4 bag skip-ack-loop.t16.p.r3, 5 bag skip-ack-loop, 5 bag correct.

left out by the table rules (timeout): 4 correct bag.

skipped: a smaller size had reached its limit: 4 skip-ack-loop fifo, 4 skip-ack-loop bag, 5 correct fifo, 5 skip-ack-loop fifo, 5 skip-ack-loop bag, 5 correct bag, 6 skip-ack-loop bag, 6 correct bag, 6 skip-ack-loop fifo, 6 correct fifo, 7 skip-ack-loop bag, 7 skip-ack-loop fifo, 7 correct fifo, 7 correct bag, 8 skip-ack-loop bag, 8 skip-ack-loop fifo, 8 correct bag, 8 correct fifo, 9 skip-ack-loop bag, 9 skip-ack-loop fifo, 9 correct bag, 9 correct fifo.

left out by the table rules (invalid: foreign load): 4 bag skip-ack-loop, 4 bag correct.

stateful

stateful: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
2	400	256	conforms	0.1	4	1 h, 250 GB	brain02, 2026-11-02
3	104976	82944	conforms	6.8	2	1 h, 250 GB	brain02, 2026-11-02
4	95883264	84934656	conforms	2010.3	222	48 h, 250 GB	brain04, 2026-11-05
6			(48 h)			48 h, 250 GB	brain04, 2026-11-03

stateful: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
2	400	256	conforms	0.1	4	1 h, 250 GB	brain02, 2026-11-02
3	104976	82944	conforms	7.7	4	1 h, 250 GB	brain02, 2026-11-02
4	95883264	84934656	conforms	2209.8	240	1 h, 250 GB	brain02, 2026-11-02
5			(1 h)			1 h, 250 GB	brain02, 2026-11-02

stateful: pair **skip-ack-loop**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	run
2	400	64	64	violation	0.1	4	1 h, 250 GB	brain02, 2026-11-02
3	104976	2304	2304	violation	3.9	4	1 h, 250 GB	brain02, 2026-11-02
4	95883264	147456	147456	violation	977.0	220	48 h, 250 GB	brain04, 2026-11-05

stateful: pair **skip-ack-loop**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	run
2	400	64	64	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
3	104976	2304	2304	violation	3.8	4	1 h, 250 GB	brain02, 2026-11-02
4	95883264	147456	147456	violation	1115.1	236	1 h, 250 GB	brain02, 2026-11-02
5				(1 h)			1 h, 250 GB	brain02, 2026-11-02

Not run (stateful). left out by the table rules (invalid: foreign load): 5 skip-ack-loop bag, 5 correct bag, 6 bag skip-ack-loop, 7 bag correct.

planned but never scheduled (grid points beyond the ladder): 5 skip-ack-loop bag, 6 correct bag, 6 skip-ack-loop bag, 7 correct bag, 7 skip-ack-loop bag, 8 correct bag, 8 skip-ack-loop bag, 9 correct bag, 9 skip-ack-loop bag, 10 correct bag, 10 skip-ack-loop bag, 11 correct bag, 11 skip-ack-loop bag.

left out by the table rules (ok): 4 skip-ack-loop bag, 4 correct bag.

left out by the table rules (timeout): 5 correct bag, 5 bag correct, 5 bag skip-ack-loop, 7 bag skip-ack-loop.

skipped: a smaller size had reached its limit: 6 correct bag, 6 skip-ack-loop fifo, 6 skip-ack-loop bag, 6 correct fifo, 7 correct fifo, 7 skip-ack-loop fifo, 7 correct bag, 7 skip-ack-loop bag, 8 skip-ack-loop fifo, 8 correct bag, 8 correct fifo, 8 skip-ack-loop bag, 9 skip-ack-loop fifo, 9 correct bag, 9 correct fifo, 9 skip-ack-loop bag.

offline

offline: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	400	256	conforms	0.1	8	1 h, 250 GB	0.1	0.1	0.0	merge	brain05, 2026-11-03
3	104976	82944	conforms	4.4	8	1 h, 250 GB	4.4	3.7	0.0	merge	brain05, 2026-11-03
4	95883264	84934656	conforms	1329.7	412	48 h, 250 GB	1329.7	1186.8	0.0	merge	brain04, 2026-11-05
4	95883264	84934656	conforms	1326.7	444	1 h, 250 GB	1326.6	1202.7	0.0	merge	brain05, 2026-11-03

offline: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	400	256	conforms	0.2	8	1 h, 250 GB	0.1	0.2	0.0	merge	brain05, 2026-11-03
3	104976	82944	conforms	4.7	8	1 h, 250 GB	4.7	3.8	0.0	merge	brain05, 2026-11-03
4	95883264	84934656	conforms	1412.9	442	1 h, 250 GB	1412.8	1192.9	0.1	merge	brain05, 2026-11-03

offline: pair **skip-ack-loop**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	400	64	16	violation	0.1	8	1 h, 250 GB	0.1	0.1	0.0	merge	brain05, 2026-11-03
3	104976	2304	64	violation	4.4	8	1 h, 250 GB	4.4	0.2	0.0	merge	brain05, 2026-11-03
4	95883264	147456	256	violation	1329.7	226	48 h, 250 GB	1329.7	6.7	0.0	merge	brain04, 2026-11-05
4	95883264	147456	256	violation	1326.7	244	1 h, 250 GB	1326.6	6.4	0.0	merge	brain05, 2026-11-03

offline: pair **skip-ack-loop**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	400	64	16	violation	0.1	8	1 h, 250 GB	0.1	0.1	0.0	merge	brain05, 2026-11-03
3	104976	2304	64	violation	4.7	8	1 h, 250 GB	4.7	0.2	0.0	merge	brain05, 2026-11-03
4	95883264	147456	256	violation	1412.9	238	1 h, 250 GB	1412.8	6.5	0.0	merge	brain05, 2026-11-03

Not run (offline). planned but never scheduled (grid points beyond the ladder): 5 bag skip-ack-loop, 5 bag correct, 8 skip-ack-loop bag.

skipped: a smaller size had reached its limit: 5 correct fifo, 5 correct bag, 6 correct bag, 6 skip-ack-loop fifo, 6 skip-ack-loop bag, 6 correct fifo, 7 correct fifo, 7 skip-ack-loop fifo, 7 correct bag, 7 skip-ack-loop bag, 8 skip-ack-loop fifo, 8 correct bag, 8 correct fifo, 8 skip-ack-loop bag, 9 skip-ack-loop fifo, 9 correct bag, 9 correct fifo, 9 skip-ack-loop bag.

refused (an input missing at run time): 5 skip-ack-loop bag, 5 skip-ack-loop fifo.

cross offline, Java implementation against the Rust specification

cross offline, Java implementation against the Rust specification: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	400	256	conforms	5.1	175	10 min, 250 GB	0.1	5.1	0.0	merge	brain01, 2026-11-03
3	104976	82944	conforms	108.2	5261	48 h, 80 GB	6.8	108.2	0.0	merge	pinky08, 2026-11-07

cross offline, Java implementation against the Rust specification: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	400	256	conforms	5.2	182	10 min, 250 GB	0.1	5.1	0.1	merge	brain01, 2026-11-03
3	104976	82944	conforms	107.6	5123	48 h, 80 GB	6.5	107.5	0.0	merge	pinky07, 2026-11-07

cross offline, Java implementation against the Rust specification: pair **skip-ack-loop**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	400	64	16	violation	7.5	155	10 min, 250 GB	0.1	7.5	0.1	merge	brain01, 2026-11-03
3	104976	2304	64	violation	10.5	2347	48 h, 80 GB	6.8	10.5	0.0	merge	pinky08, 2026-11-07

cross offline, Java implementation against the Rust specification: pair **skip-ack-loop**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	400	64	16	violation	7.8	167	10 min, 250 GB	0.1	7.7	0.1	merge	brain01, 2026-11-03
3	104976	2304	64	violation	11.4	2362	48 h, 80 GB	6.5	11.4	0.0	merge	pinky07, 2026-11-07

Not run (cross offline, Java implementation against the Rust specification). skipped: a smaller size had reached its limit: 5 bag correct, 5 bag skip-ack-loop.

planned but never scheduled (grid points beyond the ladder): 4 bag skip-ack-loop, 4 bag correct.

cross offline, Rust implementation against the Java specification

cross offline, Rust implementation against the Java specification: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	400	256	conforms	5.3	171	10 min, 250 GB	5.1	0.1	0.2	merge	brain01, 2026-11-03
3	104976	82944	conforms	155.7	10901	10 min, 250 GB	155.6	3.7	0.1	merge	brain01, 2026-11-03

cross offline, Rust implementation against the Java specification: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	400	256	conforms	5.3	181	10 min, 250 GB	5.3	0.2	0.0	merge	brain01, 2026-11-03
3	104976	82944	conforms	266.7	12242	10 min, 250 GB	266.6	3.8	0.1	merge	brain01, 2026-11-03

cross offline, Rust implementation against the Java specification: pair **skip-ack-loop**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	400	64	16	violation	5.8	171	10 min, 250 GB	5.1	0.1	0.7	merge	brain01, 2026-11-03
3	104976	2304	64	violation	155.7	10901	10 min, 250 GB	155.6	0.2	0.1	merge	brain01, 2026-11-03

cross offline, Rust implementation against the Java specification: pair **skip-ack-loop**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	400	64	16	violation	5.3	181	10 min, 250 GB	5.3	0.1	0.0	merge	brain01, 2026-11-03
3	104976	2304	64	violation	266.7	12242	10 min, 250 GB	266.6	0.2	0.0	merge	brain01, 2026-11-03

Not run (cross offline, Rust implementation against the Java specification). planned but never scheduled (grid points beyond the ladder): 4 bag skip-ack-loop, 4 bag correct, 5 bag correct, 5 bag skip-ack-loop.

cross online, Rust implementation (TraceForge) against the Java specification export

cross online, Rust implementation (TraceForge) against the Java specification export: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	400	256	conforms	4.2	229	1 h, 250 GB		brain01, 2026-11-02

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
3	104976	82944	conforms	211.9	14692	1 h, 250 GB		brain01, 2026-11-02

cross online, Rust implementation (TraceForge) against the Java specification export: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	400	256	conforms	4.4	200	1 h, 250 GB		brain01, 2026-11-02
3	104976	82944	conforms	218.9	15291	1 h, 250 GB		brain01, 2026-11-02

cross online, Rust implementation (TraceForge) against the Java specification export: pair **skip-ack-loop**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2	400	64	64	violation	4.2	229	1 h, 250 GB	2	brain01, 2026-11-02
3	104976	2304	2304	violation	207.0	14692	1 h, 250 GB	3	brain01, 2026-11-02

cross online, Rust implementation (TraceForge) against the Java specification export: pair **skip-ack-loop**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2	400	64	64	violation	4.4	200	1 h, 250 GB	2	brain01, 2026-11-02
3	104976	2304	2304	violation	214.0	15291	1 h, 250 GB	2	brain01, 2026-11-02

Not run (cross online, Rust implementation (TraceForge) against the Java specification export). planned but never scheduled (grid points beyond the ladder): 4 bag skip-ack-loop, 4 bag correct, 5 bag correct, 5 bag skip-ack-loop.

twins offline, Rust implementation against its Java twin

twins offline, Rust implementation against its Java twin: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	256	256	conforms	5.1	175	10 min, 250 GB	5.1	0.1	0.0	merge	brain01, 2026-11-03
3	82944	82944	conforms	139.1	9257	10 min, 250 GB	139.0	3.7	0.1	merge	brain01, 2026-11-03

twins offline, Rust implementation against its Java twin: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	256	256	conforms	5.1	182	10 min, 250 GB	5.1	0.2	0.0	merge	brain01, 2026-11-03
3	82944	82944	conforms	265.9	10023	10 min, 250 GB	265.9	3.8	0.0	merge	brain01, 2026-11-03

twins offline, Rust implementation against its Java twin: pair **skip-ack-loop**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	64	64	conforms	7.7	155	10 min, 250 GB	7.5	0.1	0.2	merge	brain01, 2026-11-03
3	2304	2304	conforms	18.8	2191	10 min, 250 GB	18.7	0.2	0.0	merge	brain01, 2026-11-03

twins offline, Rust implementation against its Java twin: pair **skip-ack-loop**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	64	64	conforms	7.7	167	10 min, 250 GB	7.7	0.1	0.0	merge	brain01, 2026-11-03
3	2304	2304	conforms	10.0	576	10 min, 250 GB	10.0	0.2	0.0	merge	brain01, 2026-11-03

twins offline, Rust implementation against its Java twin: pair **skip-ack-loop**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
4	147456	147456		conforms*	202.5	9300	48 h, 200 GB	202.4	8.6	0.0	merge	volta13, 2026-11-07

twins offline, Rust implementation against its Java twin: pair **skip-ack-loop**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	256	64	16	violation	5.1	175	10 min, 250 GB	5.1	0.1	0.0	merge	brain01, 2026-11-03
3	82944	2304	64	violation	139.2	9257	10 min, 250 GB	139.0	0.2	0.2	merge	brain01, 2026-11-03

twins offline, Rust implementation against its Java twin: pair **skip-ack-loop**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	256	64	16	violation	5.2	182	10 min, 250 GB	5.1	0.1	0.1	merge	brain01, 2026-11-03
3	82944	2304	64	violation	266.2	10023	10 min, 250 GB	265.9	0.2	0.3	merge	brain01, 2026-11-03

Not run (twins offline, Rust implementation against its Java twin). skipped: a smaller size had reached its limit: 4 fifo skip-ack-loop.

planned but never scheduled (grid points beyond the ladder): 4 bag skip-ack-loop, 4 bag correct.

twins offline, Java implementation against its Rust twin

twins offline, Java implementation against its Rust twin: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	256	256	conforms	5.1	175	10 min, 250 GB	0.1	5.1	0.0	merge	brain01, 2026-11-03
3	82944	82944	conforms	139.2	9257	10 min, 250 GB	3.7	139.0	0.2	merge	brain01, 2026-11-03

twins offline, Java implementation against its Rust twin: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	256	256	conforms	5.1	182	10 min, 250 GB	0.2	5.1	0.0	merge	brain01, 2026-11-03
3	82944	82944	conforms	266.5	10023	10 min, 250 GB	3.8	265.9	0.6	merge	brain01, 2026-11-03

twins offline, Java implementation against its Rust twin: pair **skip-ack-loop**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	64	64	conforms	7.5	155	10 min, 250 GB	0.1	7.5	0.0	merge	brain01, 2026-11-03
3	2304	2304	conforms	19.2	2191	10 min, 250 GB	0.2	18.7	0.5	merge	brain01, 2026-11-03

twins offline, Java implementation against its Rust twin: pair **skip-ack-loop**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	64	64	conforms	7.7	167	10 min, 250 GB	0.1	7.7	0.0	merge	brain01, 2026-11-03
3	2304	2304	conforms	10.2	576	10 min, 250 GB	0.2	10.0	0.2	merge	brain01, 2026-11-03

twins offline, Java implementation against its Rust twin: pair **skip-ack-loop**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
4	147456	147456		conforms*	202.5	9300	48 h, 200 GB	8.6	202.4	0.0	merge	volta13, 2026-11-07
4	84934656	147456	256	violation	1685.9	9511	48 h, 200 GB	1685.9	202.4	0.0	merge	volta13, 2026-11-07

twins offline, Java implementation against its Rust twin: pair **skip-ack-loop**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	256	64	16	violation	7.5	155	10 min, 250 GB	0.1	7.5	0.0	merge	brain01, 2026-11-03
3	82944	2304	64	violation	18.9	2191	10 min, 250 GB	3.7	18.7	0.1	merge	brain01, 2026-11-03

twins offline, Java implementation against its Rust twin: pair **skip-ack-loop**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	256	64	16	violation	7.7	167	10 min, 250 GB	0.2	7.7	0.0	merge	brain01, 2026-11-03
3	82944	2304	64	violation	10.1	576	10 min, 250 GB	3.8	10.0	0.1	merge	brain01, 2026-11-03

Not run (twins offline, Java implementation against its Rust twin). skipped: a smaller size had reached its limit: 4 fifo skip-ack-loop.

planned but never scheduled (grid points beyond the ladder): 4 bag correct.

context-free twins offline, Rust implementation against the context-free Java port

context-free twins offline, Rust implementation against the context-free Java port: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	256	256	conforms	7.1	162	10 min, 250 GB	7.0	0.1	0.1	merge	brain01, 2026-11-03
3	82944	82944	conforms	142.6	9072	10 min, 250 GB	142.5	3.7	0.1	merge	brain01, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	256	256	conforms	6.2	180	10 min, 250 GB	6.2	0.2	0.0	merge	brain01, 2026-11-03

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
3	82944	82944	conforms	161.1	9823	10 min, 250 GB	160.8	3.8	0.3	merge	brain01, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair **skip-ack-loop**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	64	64	conforms	4.3	160	10 min, 250 GB	4.3	0.1	0.0	merge	brain01, 2026-11-03
3	2304	2304	conforms	11.7	2948	10 min, 250 GB	11.7	0.2	0.0	merge	brain01, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair **skip-ack-loop**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	64	64	conforms	7.2	160	10 min, 250 GB	7.1	0.1	0.1	merge	brain01, 2026-11-03
3	2304	2304	conforms	12.2	2893	10 min, 250 GB	12.1	0.2	0.1	merge	brain01, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair **skip-ack-loop**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
4	147456	147456		conforms*	210.2	9587	48 h, 200 GB	210.2	8.6	0.0	merge	volta13, 2026-11-07

context-free twins offline, Rust implementation against the context-free Java port: pair **skip-ack-loop**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	256	64	16	violation	7.2	163	10 min, 250 GB	7.0	0.1	0.3	merge	brain01, 2026-11-03
3	82944	2304	64	violation	142.6	9072	10 min, 250 GB	142.5	0.2	0.0	merge	brain01, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair **skip-ack-loop**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	256	64	16	violation	6.2	180	10 min, 250 GB	6.2	0.1	0.0	merge	brain01, 2026-11-03
3	82944	2304	64	violation	160.8	9823	10 min, 250 GB	160.8	0.2	0.0	merge	brain01, 2026-11-03

Not run (context-free twins offline, Rust implementation against the context-free Java port). skipped: a smaller size had reached its limit: 4 fifo skip-ack-loop.

planned but never scheduled (grid points beyond the ladder): 4 bag skip-ack-loop, 4 bag correct.

context-free twins offline, Java port against the Rust implementation

context-free twins offline, Java port against the Rust implementation: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	256	256	conforms	7.0	162	10 min, 250 GB	0.1	7.0	0.0	merge	brain01, 2026-11-03
3	82944	82944	conforms	142.5	9072	10 min, 250 GB	3.7	142.5	0.0	merge	brain01, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	256	256	conforms	6.2	180	10 min, 250 GB	0.2	6.2	0.0	merge	brain01, 2026-11-03
3	82944	82944	conforms	160.8	9823	10 min, 250 GB	3.8	160.8	0.1	merge	brain01, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair **skip-ack-loop**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	64	64	conforms	4.4	160	10 min, 250 GB	0.1	4.3	0.2	merge	brain01, 2026-11-03
3	2304	2304	conforms	11.7	2948	10 min, 250 GB	0.2	11.7	0.0	merge	brain01, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair **skip-ack-loop**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	64	64	conforms	7.1	160	10 min, 250 GB	0.1	7.1	0.0	merge	brain01, 2026-11-03
3	2304	2304	conforms	12.2	2893	10 min, 250 GB	0.2	12.1	0.0	merge	brain01, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair **skip-ack-loop**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	256	64	16	violation	4.4	160	10 min, 250 GB	0.1	4.3	0.1	merge	brain01, 2026-11-03
3	82944	2304	64	violation	11.7	2948	10 min, 250 GB	3.7	11.7	0.0	merge	brain01, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair **skip-ack-loop**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	256	64	16	violation	7.1	160	10 min, 250 GB	0.2	7.1	0.0	merge	brain01, 2026-11-03
3	82944	2304	64	violation	12.2	2893	10 min, 250 GB	3.8	12.1	0.0	merge	brain01, 2026-11-03

Not run (context-free twins offline, Java port against the Rust implementation). skipped: a smaller size had reached its limit: 4 fifo skip-ack-loop.

planned but never scheduled (grid points beyond the ladder): 4 bag correct.

refused (an input missing at run time): 4 bag skip-ack-loop.

context-free twins online, Java port (JMC) against the Rust implementation's export

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	256	256	conforms	8.7	174	1 h, 250 GB		brain01, 2026-11-03
3	82944	82944	conforms	192.4	10872	1 h, 250 GB		brain01, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	256	256	conforms	8.4	168	1 h, 250 GB		brain01, 2026-11-03
3	82944	82944	conforms	211.1	11897	1 h, 250 GB		brain01, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **skip-ack-loop**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	64	64	conforms	8.6	166	1 h, 250 GB		brain01, 2026-11-03
3	2304	2304	conforms	16.7	2602	1 h, 250 GB		brain01, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **skip-ack-loop**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	64	64	conforms	8.1	171	1 h, 250 GB		brain01, 2026-11-03
3	2304	2304	conforms	17.6	2482	1 h, 250 GB		brain01, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **skip-ack-loop**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2	256	64	64	violation	7.5	178	1 h, 250 GB	192	brain01, 2026-11-03
3	82944	2304	2304	violation	21.4	2491	1 h, 250 GB	312	brain01, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **skip-ack-loop**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2	256	64	64	violation	7.7	165	1 h, 250 GB	168	brain01, 2026-11-03
3	82944	2304	2304	violation	21.8	2445	1 h, 250 GB	199	brain01, 2026-11-03

Not run (context-free twins online, Java port (JMC) against the Rust implementation's export). planned but never scheduled (grid points beyond the ladder): 4 bag skip-ack-loop, 4 bag correct.

context-free twins online, Rust implementation (TraceForge) against the Java port's export

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	256	256	conforms	4.0	205	1 h, 250 GB		brain01, 2026-11-02
3	82944	82944	conforms	177.5	12510	1 h, 250 GB		brain01, 2026-11-02

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	256	256	conforms	4.0	196	1 h, 250 GB		brain01, 2026-11-02
3	82944	82944	conforms	207.9	11112	1 h, 250 GB		brain01, 2026-11-02

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair skip-ack-loop, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	64	64	conforms	3.4	158	1 h, 250 GB		brain01, 2026-11-02
3	2304	2304	conforms	10.2	762	1 h, 250 GB		brain01, 2026-11-02

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair skip-ack-loop, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	64	64	conforms	3.4	154	1 h, 250 GB		brain01, 2026-11-02
3	2304	2304	conforms	10.8	3697	1 h, 250 GB		brain01, 2026-11-02

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair skip-ack-loop, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2	256	64	64	violation	4.0	205	1 h, 250 GB	2	brain01, 2026-11-02
3	82944	2304	2304	violation	172.6	12510	1 h, 250 GB	2	brain01, 2026-11-02

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair skip-ack-loop, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2	256	64	64	violation	4.0	196	1 h, 250 GB	2	brain01, 2026-11-02
3	82944	2304	2304	violation	203.1	11112	1 h, 250 GB	2	brain01, 2026-11-02

Not run (context-free twins online, Rust implementation (TraceForge) against the Java port's export). planned but never scheduled (grid points beyond the ladder): 4 bag skip-ack-loop, 4 bag correct.